

Fixation Plane as an Attentional Plane

Background: the Far > Near asymmetry

The DecoupledDots and DepthColorLinked experiments consistently produce a Far > Near performance asymmetry of approximately 15 pp (DecoupledDots GLM F4 AME: -15.3 pp for Near; $p < .001$). The translating field is identified more reliably when it occupies the Far depth plane than the Near depth plane, regardless of swap condition. This asymmetry is absent under monocular viewing, establishing a binocular/stereoscopic origin. The mechanism, however, is unresolved.

Prior account: the onset-cued plane as attentional anchor

Prior discussion focused on the onset-cued surface as the primary attentional plane. The delayed-onset field captures exogenous attention, establishing a depth representation that the observer tracks across the cue interval. By analogy with the ‘attended object extends attention behind it’ principle, the cued plane could preferentially propagate attentional weight toward objects at greater depth—i.e., toward Far when the cued field occupies the Near plane, and further beyond Far when the cued field is in the Far plane. This cue-locked account predicts that the Far > Near gap should be larger in the CUED arm, and possibly absent or reversed in the UNCUED arm.

New hypothesis: the fixation plane as attentional anchor

The present note evaluates an additional, non-mutually-exclusive hypothesis: the fixation plane itself functions as a persistent attentional depth anchor. In stereoscopic VR, the fixation cross is rendered at a specific virtual depth. The two dot fields flank this depth symmetrically in disparity magnitude: the Near field carries crossed disparity, the Far field carries uncrossed disparity. If attentional weight extends preferentially behind the fixation plane—just as it may extend behind the cued object’s plane—the Far field should receive a standing attentional advantage that is present on every trial, independent of which field was cued by temporal onset. This would manifest as a Far > Near advantage in the UNCUED arm as well as the CUED arm.

Compatibility of the two mechanisms

The two mechanisms are compatible and potentially additive. The onset-cue mechanism is trial-by-trial and cue-locked; the fixation mechanism is structural and present throughout every trial. When the Far field happens to be cued, both mechanisms favour the same field. When the Near field is cued, the onset-cue mechanism favours Near while the fixation mechanism continues to pull toward Far, creating partial opposition.

Onset-cue mechanism: cue-locked → attentional plane = cued field’s depth → weight extends behind cued field

Fixation-plane mechanism: structural → attentional plane = fixation depth → weight extends toward Far (every trial)

Combined (Far cued): both mechanisms favour Far → maximum advantage

Combined (Near cued): onset-cue favours Near; fixation-plane favours Far → partial opposition

Predictions and Existing Data

What each account predicts and where existing data bears on it

Table 1. Predictions of the two accounts and what existing data can address.

Prediction	Cue-locked account	Fixation-plane account	Existing data
Far > Near in UNCUEd arm	No (cue absent)	Yes (fixation bias always on)	Check cell means → diagnostic
Far > Near larger in CUED arm than UNCUEd arm	Yes (cue amplifies)	Partial (cue adds to standing bias)	F1×F4 interaction in GLM → diagnostic
Asymmetry grows with depth separation	Ambiguous	Yes if bias is signal-driven	DepthParam data → consistent
Asymmetry absent monocularly	Yes (no disparity for depth repr.)	Yes (no crossed/uncrossed disparity)	Confirmed
Asymmetry shifts when fixation depth moved	No (onset-cue determines plane)	Yes (direction tracks fixation)	Not yet collected
At very short SOA, Far > Near persists (before cue matures)	No (cue not yet established)	Yes (fixation bias present from t=0)	SOA experiment pending

Interpretation

The critical diagnostic in existing data is the UNCUEd arm. Under a purely cue-locked account, the UNCUEd arm should show no Far > Near effect: without an onset cue, there is no attentional depth plane to extend from. If Far > Near is present in the UNCUEd trials—regardless of swap condition—that is positive evidence for a standing structural bias, consistent with the fixation-plane hypothesis (or with a sensory confound; see p. 3). The F1 × F4 interaction term in the GLM addresses the second row of the table: if the interaction is significant and positive, the Far advantage is amplified by the onset cue, suggesting both mechanisms contribute. If F1 × F4 is near zero while the F4 main effect is large, the fixation-plane account needs no additional cue-locked component.

Complications and Mechanism Alternatives

Sensory confounds, neural-population account, interaction structure

1 — Vergence-comfort confound

A sensory confound must be considered before attributing the Far > Near gap to attentional depth weighting. Crossed disparity (Near) and uncrossed disparity (Far) are not perceptually symmetric. Fusional vergence is known to be faster and more accurate for small uncrossed disparities than for matched crossed disparities, particularly under sustained viewing. In VR, the vergence-accommodation conflict adds further asymmetry: the display is at a fixed optical distance while vergence tracks virtual depth. Objects rendered with uncrossed disparity (Far) require less additional vergence effort relative to the resting fixation distance, and may be more stably fused. If the Far plane is simply cleaner stereoscopically, motion direction detection would be easier there without any attentional mechanism. This confound is difficult to rule out entirely, but the fact that the asymmetry ‘grows’ with depth separation—rather than flattening as perceptual fusion becomes more difficult at larger disparities—is at least weakly inconsistent with a simple sensory-quality account.

2 — Neural-population asymmetry (sensory, not attentional)

At the neural level, near-zero and far-tuned disparity neurons are not distributed symmetrically in early visual areas. In macaque V1 and MT, the population of far-tuned (uncrossed-disparity) cells is somewhat larger and shows higher signal-to-noise than near-tuned cells at moderate disparities. If the Far dot field drives a larger or more reliable population response in motion-sensitive areas, the downstream readout for direction discrimination would be more accurate independently of attention. This is a neural-population account rather than a vergence-comfort account, but both are sensory rather than attentional. Neither can be ruled out with the current data.

3 — Additive structure of the two attentional mechanisms

The additive structure expected if both mechanisms operate simultaneously is: CUED · Far is optimal (onset-cue + fixation bias both favour Far), CUED · Near and UNCUED · Far occupy intermediate positions (one mechanism favours, one does not), and UNCUED · Near is worst (neither mechanism favours). A purely cue-locked account with no fixation mechanism predicts $\text{UNCUED} \cdot \text{Far} \approx \text{UNCUED} \cdot \text{Near}$ (chance-level and equal). If the observed UNCUED cells show a Far advantage, that ordering directly falsifies the purely cue-locked account.

4 — Possible opposing interaction

It is also worth noting the possibility of an opposing interaction. If the fixation plane’s attentional bias toward Far is strong, then when the Near field is cued by onset, the fixation mechanism actively works against the trial’s informative cue. This would predict that CUED · Near should be disproportionately impaired relative to a simple additive prediction. In other words, the onset cue’s benefit for Near should be smaller than its benefit for Far—exactly a positive F1 × F4 interaction. Checking whether the observed F1 × F4 coefficient is positive, negative, or near zero is therefore the primary

Critical Experiments and Summary

What to do next to test the fixation-plane hypothesis

1 — Check UNCUED Near vs. Far in existing data [can do now]

Split existing DecoupledDots and DepthColorLinked data by F4 within the UNCUED arm only. If Far > Near is present there (ideally collapsing across swap conditions to maximise n), that is the first positive evidence for a standing fixation bias. The F1 × F4 interaction in the GLM provides the complementary evidence that the fixation bias and onset-cue mechanism interact.

2 — Fixation-depth manipulation [new experiment]

Render the fixation cross at the Near plane depth for one block and at the Far plane depth for another. Everything else (dot fields, depth separation, swap conditions) is unchanged. Fixation-plane hypothesis predicts: Far > Near advantage should weaken when fixation is moved to Far depth (the fixation plane is no longer "behind" the Far field), and possibly reverse when fixation is moved beyond Far (Far is now the "near" side of fixation). This is the cleanest test. Can be added as an extra condition within a DepthSwapCtrl session (no new experiment spec required, just a fixation depth parameter).

3 — SOA manipulation [already planned]

Vary the delay between Field B onset and translation onset. At very short SOAs the onset-cue attentional representation has not matured; the CUED · Near advantage should be small. But if a fixation-based Far bias is structural, the Far > Near gap in the UNCUED arm should be flat across SOA, while the CUED arm shows an SOA-dependent component. Crossing SOA with Far/Near plane would decompose the two contributions.

4 — Zero-disparity baseline [already noted as needed]

A no-depth session (disparity = 0 for both fields, rendered at fixation plane) would establish the flat baseline. Any Far > Near gap in the stereoscopic conditions is then attributable to the depth manipulation specifically. Without this anchor it remains ambiguous whether Far is advantaged or Near is penalised. The fixation-plane hypothesis predicts that as soon as disparity is introduced, performance splits: Far rises above baseline, Near falls below, or both. The monocular sessions are the closest available proxy for this baseline; they show the gap collapses under monocular viewing, consistent with a disparity-dependent mechanism.

Summary

The fixation-plane hypothesis offers a parsimonious account of the Far > Near asymmetry as a standing structural bias, present on every trial and reflecting either attentional weighting toward objects behind the fixation plane, a physiological asymmetry in uncrossed-vs.-crossed disparity processing, or a vergence-comfort advantage for Far. It is compatible with—and potentially additive with—the onset-cue attentional plane mechanism.

The key distinction is whether the asymmetry is present in the UNCUED arm (fixation or sensory) or only in the CUED arm (cue-locked). This can be checked in existing data immediately. The fixation-depth manipulation experiment would provide the cleanest causal test.